RASHMI PANDEY

Software Development Engineer (SDE) Candidate • Core Java, AWS & Distributed Systems Focus

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EDUCATION

Amity University Jharkhand — Bachelor of Technology (B.Tech) in Computer Science & Engineering `2024 – 2028`

CGPA: 8.00 / 10.0 • Senior Secondary (Class XII): 88.2% • Secondary (Class X): 85.0%

TECHNICAL SKILLS

Languages:	Java (Core Java, Multithreading, OOP, Java 17), SQL (MySQL, T-SQL), Python 3, JavaScript (ES6+), C/C++
Backend & Architecture:	Jakarta EE (Servlets 6.0, DAO Pattern, JDBC), RESTful APIs, Node.js, Express, Tomcat 11, Concurrency
Databases & Systems:	MySQL (InnoDB, B-Tree Indexing, ACID Transactions), Azure SQL, MongoDB Atlas, Redis (Caching)
Cloud & DevOps:	AWS (EC2, CloudWatch, S3, IAM, Lambda), Microsoft Azure, Docker, Linux/Unix Shell, Git, Maven, PM2
Core Competencies:	Data Structures & Algorithms, Distributed Systems, Database Internals, Latency Optimization, Object-Oriented Design

RESEARCH & ENGINEERING EXPERIENCE

National Institute of Advanced Manufacturing Technology (NIAMT), Ranchi `Jun 2026 – Jul 2026`

Summer Research Intern — Distributed Systems & Cloud Infrastructure (Guided by Dr. Rahul Mishra)

- Architected **DVQ**, an intelligent carbon-aware workload router evaluated across 7 AWS regional grid topologies, formulating request placement under Lyapunov optimization to balance strict latency SLAs against carbon intensity.
- Achieved carbon footprint reductions of **35.87% (Interactive)**, **30.28% (Realtime)**, and **16.58% (Batch)** versus Least-Connections baseline, with a verified **0.0% SLA violation rate** across all evaluation trials.
- Maintained a median routing decision overhead of **sub-100µs** under Linux PM2 supervision; evaluated system resilience under synthetic chaos tests with simulated regional AWS grid outages and traffic surges.

KEY ENGINEERING PROJECTS

EXAMORA — Enterprise Academic Testing & Evaluation Platform (Java 17, Jakarta EE, Tomcat 11, MySQL) `github.com/rashmipandey870/EXAMORA`

- Constructed a high-throughput examination portal utilizing native **Jakarta Servlets 6.0** with the **Data Access Object (DAO)** pattern over raw JDBC connection pooling, eliminating framework runtime overhead.
- Engineered an **ACID-compliant relational MySQL schema** across 10+ staged migrations; implemented composite B-Tree indexes on (exam_id, student_id) and row-level locks to guarantee data integrity during concurrent exam submissions.
- Implemented BCrypt cryptographic password hashing (salt rounds = 12) for secure authentication; built an ingestion pipeline with Apache PDFBox to parse and index multi-page question bank documents.

Sangathan — Algorithmic Tournament Scheduling Platform (Android Java, Node.js, Express, MongoDB) `github.com/rashmipandey870/Sangathan-...`

- Developed an automated tournament management system generating collision-free cyclic round-robin (Berger Algorithm) and knockout bracket graphs for 100+ athletes across 6 simultaneous sports.
- Built 10+ RESTful backend endpoints in Node.js/Express (github.com/rashmipandey870/Sangathan-Backend) with indexed MongoDB queries, delivering sub-150ms response times and robust state synchronization via Android Retrofit 2.

Study Planner — PriorityQueue Min-Heap Algorithmic Scheduler (Java 17, Data Structures & Heuristics) `github.com/rashmipandey870/Study-Planner-...`

- Implemented an O(N log N) min-heap scheduling engine that dynamically recalculates priority order balancing task urgency deadlines and cognitive fatigue decay models with sub-millisecond overhead.

CERTIFICATIONS & ALGORITHMIC ACHIEVEMENTS

• **Microsoft Certified: Azure Database Administrator Associate** (Credential ID: **647C0C2C5B35FA32**) — Demonstrated mastery in relational query execution plans, B-Tree index optimization, transaction locking, and high availability.

• **NPTEL Elite + Silver (Top 2% Topper): Python for Data Science** — **IIT Madras** (Score: **84%**).

• **NPTEL Elite + Silver: The Joy of Computing using Python** — **IIT Ropar** (Score: **87%**).

• **100 Days of Code (Data Structures & Algorithms):** Solved **175+ curated algorithmic problems** on LeetCode & GitHub (github.com/rashmipandey870/100-days-of-code) covering Trees, Graphs (Dijkstra, BFS/DFS), Dynamic Programming, and Heaps. Active GitHub history with **1,200+ commits**.